

Supplementary Material for “Backpropagation-Free Test-Time Adaptation for Lightweight EEG-Based Brain-Computer Interfaces”

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S-I. THEORETICAL FOUNDATION FOR BFT

This section of the Supplementary Material gives a variance-based justification for BFT. It models transformations as draws from a design distribution and shows the conditions under which aggregating their predictions reduces variance. For deterministic BFT-D, the fixed mask bank is one realized design whose branch identities are retained at deployment, and no per-trial mask resampling is implied. Variance reduction yields more stable outputs but does not by itself guarantee higher accuracy under every domain shift.

A. Label-Preserving Test-Time Randomization

Definition 1 (Test-Time Randomization and Aggregation). *Let ζ index a label-preserving member of the transformation design distribution (e.g., a BFT-A transformation or a candidate structured feature mask). Fix an input $\mathbf{x}$. Define the scalar prediction under ζ as*

$$f(\zeta; \mathbf{x}) := \mathbb{E}[y|\mathbf{x}; \zeta] \in \mathbb{R}, \quad (\text{S1})$$

where the expectation is taken with respect to the model-induced predictive distribution. In the notation of the main text, $f(\zeta_k; \mathbf{x})$ corresponds to the prediction $h(\mathbf{z}^{(k)})$ of the k -th transformed input, and the weights $\mathbf{w}$ introduced below are those produced by the ranking module $r(\cdot)$.

To quantify prediction variability across the transformation design, single-branch uncertainty is measured via the variance:

$$V_0 := \text{Var}_\zeta(f(\zeta; \mathbf{x})). \quad (\text{S2})$$

Let $\zeta_1, \dots, \zeta_K$ denote the selected design members and set $f_k := f(\zeta_k; \mathbf{x})$. The k -th member defines the k -th test-time branch. For analysis, the branch design may be treated as independently and identically distributed (i.i.d.). More generally, we allow $\zeta_k \sim \mathcal{A}_k$ with branch-specific distributions $\{\mathcal{A}_k\}_{k=1}^K$. BFT-D fixes the resulting bank before ranking-module training and uses the same bank at test time.

A learning-to-rank module outputs weights $\mathbf{w} = (w_1, \dots, w_K)$ with $w_k \geq 0$ and $\sum_{k=1}^K w_k = 1$, and the aggregation is

$$\hat{f}_{\mathbf{w}}(\mathbf{x}) := \sum_{k=1}^K w_k f_k. \quad (\text{S3})$$

The following deduction treats the realized weight vector $\mathbf{w}$ as deterministic. All variances are with respect to the transformation-design model $(\zeta_1, \dots, \zeta_K)$, not repeated random masks during deployed BFT-D inference.

B. Variance Decomposition for Weighted Aggregation

Lemma 1 (Exact Variance Decomposition). *Fix an input $\mathbf{x}$. Let $f_k := f(\zeta_k; \mathbf{x})$ be square-integrable random variables induced by the joint test-time randomness, and define $\mu_k := \mathbb{E}[f_k]$ for $k = 1, \dots, K$. Let $\hat{f}_{\mathbf{w}} := \sum_{k=1}^K w_k f_k$ with deterministic weights $\mathbf{w}$. Then we obtain*

$$\text{Var}(\hat{f}_{\mathbf{w}}) = \sum_{k=1}^K w_k^2 \text{Var}(f_k) + \sum_{i \neq j} w_i w_j \text{Cov}(f_i, f_j). \quad (\text{S4})$$

Proof.

$$\begin{aligned} \text{Var}(\hat{f}_{\mathbf{w}}) &= \mathbb{E} \left[\left(\hat{f}_{\mathbf{w}} - \mathbb{E}[\hat{f}_{\mathbf{w}}] \right)^2 \right] = \mathbb{E} \left[\left(\sum_{k=1}^K w_k (f_k - \mu_k) \right)^2 \right] \\ &= \sum_{i=1}^K \sum_{j=1}^K w_i w_j \mathbb{E}[(f_i - \mu_i)(f_j - \mu_j)] \\ &= \sum_{i=1}^K \sum_{j=1}^K w_i w_j \text{Cov}(f_i, f_j), \end{aligned}$$

which yields (S4) by separating diagonal and off-diagonal terms. ■

C. Homogeneous Variance Case

Assumption 1 (Homogeneous Prediction Variance). *Assume*

$$\text{Var}(f_k) = \sigma^2, \quad k = 1, \dots, K. \quad (\text{S5})$$

If $\sigma^2 = 0$, the prediction is deterministic and variance reduction is trivial. Otherwise, the correlations defined below are well-defined.

Theorem 1 (Uncertainty Reduction under Homogeneous Variance). *Define $\rho_{ij} := \text{Corr}(f_i, f_j)$ for $i \neq j$. Under Assumption 1,*

$$\text{Var}(\hat{f}_{\mathbf{w}}(\mathbf{x})) = \sigma^2 \sum_{k=1}^K w_k^2 + \sigma^2 \sum_{i \neq j} w_i w_j \rho_{ij}. \quad (\text{S6})$$

Let

$$\rho_{\max} := \max_{i \neq j, i, j \in \{1, \dots, K\}} |\rho_{ij}| \in [0, 1]. \quad (\text{S7})$$

Then

$$\text{Var}(\hat{f}_{\mathbf{w}}(\mathbf{x})) \leq \sigma^2 \left(\rho_{\max} + (1 - \rho_{\max}) \sum_{k=1}^K w_k^2 \right) \leq \sigma^2. \quad (\text{S8})$$

Moreover, $\text{Var}(\hat{f}_{\mathbf{w}}(\mathbf{x})) < \sigma^2$ whenever $\rho_{\max} < 1$ and $\sum_{k=1}^K w_k^2 < 1$.

Proof. By Lemma 1 and Assumption 1, $\text{Cov}(f_i, f_j) = \rho_{ij}\sigma^2$ for $i \neq j$, which gives (S6). For the bound, use $|\rho_{ij}| \leq \rho_{\max}$ and $\sum_{i \neq j} w_i w_j = 1 - \sum_k w_k^2$:

$$\sum_{i \neq j} w_i w_j \rho_{ij} \leq \sum_{i \neq j} w_i w_j |\rho_{ij}| \leq \rho_{\max} \left(1 - \sum_{k=1}^K w_k^2\right),$$

where we used $(\sum_k w_k)^2 = 1 = \sum_k w_k^2 + \sum_{i \neq j} w_i w_j$. Substituting into (S6) yields (S8). ■

An illustrative visualization of Theorem 1 is provided in Fig. S1.

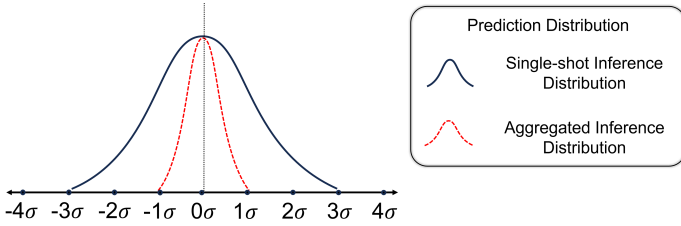


Fig. S1. Illustration of uncertainty reduction achieved through test-time transformations under the homogeneous variance assumption.

D. Heterogeneous Variance Case

In the heterogeneous-variance setting, the ensemble uncertainty is mainly affected by three factors: the worst-branch noise κ , the similarity between branches $\rho_{\max}$, and how spread the weights are K_{eff} .

Assumption 2 (Heterogeneous Prediction Variance). Fix an input $\mathbf{x}$. In the k -th branch (defined in Definition 1), the prediction f_k has heterogeneous variance:

$$\text{Var}(f_k) = \sigma_k^2, \quad 0 < \sigma_k^2 \leq \sigma_{\max}^2 < \infty, \quad k = 1, \dots, K. \quad (\text{S9})$$

Let $V_0 := \text{Var}_{\zeta}(f(\zeta; \mathbf{x}))$ be the single-shot test-time variance defined in (S2), where ζ follows the randomization used in single-shot inference. Assume that the worst branch variance is controlled relative to V_0 :

$$\sigma_{\max}^2 \leq \kappa V_0, \quad 1 \leq \kappa. \quad (\text{S10})$$

Theorem 2 (Uncertainty Reduction under Heterogeneous Variance). Let $\rho_{\max}$ be defined in (S7). Under Assumption 2, for any probability weights $\mathbf{w}$ (i.e., $w_k \geq 0$ and $\sum_{k=1}^K w_k = 1$),

$$\begin{aligned} \text{Var}(\hat{f}_{\mathbf{w}}(\mathbf{x})) &\leq \sigma_{\max}^2 \left(\rho_{\max} + (1 - \rho_{\max}) \sum_{k=1}^K w_k^2 \right) \\ &\leq \kappa V_0 \left(\rho_{\max} + (1 - \rho_{\max}) \sum_{k=1}^K w_k^2 \right). \end{aligned} \quad (\text{S11})$$

Define the effective number of branches

$$K_{\text{eff}} := \frac{1}{\sum_{k=1}^K w_k^2} \in [1, K]. \quad (\text{S12})$$

If $\rho_{\max} < 1/\kappa$, then a sufficient condition for $\text{Var}(\hat{f}_{\mathbf{w}}(\mathbf{x})) < V_0$ is

$$K_{\text{eff}} > \frac{\kappa(1 - \rho_{\max})}{1 - \kappa\rho_{\max}}. \quad (\text{S13})$$

Proof. For $i \neq j$, let $\rho_{ij} := \text{Corr}(f_i, f_j)$. Then

$$\begin{aligned} |\text{Cov}(f_i, f_j)| &= |\rho_{ij}| \sqrt{\text{Var}(f_i) \text{Var}(f_j)} \\ &\leq \rho_{\max} \sqrt{\sigma_i^2 \sigma_j^2} \leq \rho_{\max} \sigma_{\max}^2. \end{aligned} \quad (\text{S14})$$

Since $w_i w_j \geq 0$,

$$\begin{aligned} \sum_{i \neq j} w_i w_j \text{Cov}(f_i, f_j) &\leq \rho_{\max} \sigma_{\max}^2 \sum_{i \neq j} w_i w_j \\ &= \rho_{\max} \sigma_{\max}^2 \left(1 - \sum_{k=1}^K w_k^2\right), \end{aligned} \quad (\text{S15})$$

where we used $(\sum_k w_k)^2 = 1 = \sum_k w_k^2 + \sum_{i \neq j} w_i w_j$. Moreover,

$$\sum_{k=1}^K w_k^2 \text{Var}(f_k) = \sum_{k=1}^K w_k^2 \sigma_k^2 \leq \sigma_{\max}^2 \sum_{k=1}^K w_k^2. \quad (\text{S16})$$

Combining (S15) and (S16) with (S4) yields the first inequality in (S11). The second inequality follows from (S10).

For the sufficient condition, it is enough to ensure that the upper bound in (S11) is strictly smaller than V_0 , namely

$$\kappa \left(\rho_{\max} + (1 - \rho_{\max}) \sum_{k=1}^K w_k^2 \right) < 1. \quad (\text{S17})$$

This inequality requires $1 - \kappa\rho_{\max} > 0$, i.e., $\rho_{\max} < 1/\kappa$, and under this condition it is equivalent to

$$\sum_{k=1}^K w_k^2 < \frac{1 - \kappa\rho_{\max}}{\kappa(1 - \rho_{\max})}. \quad (\text{S18})$$

Using $K_{\text{eff}} = 1/\sum_{k=1}^K w_k^2$, we obtain (S13). ■

The bound improves when branches are less correlated (small $\rho_{\max}$) and the weights are not overly concentrated (large K_{eff}). These are sufficient conditions, not properties guaranteed by the architecture. They also identify failure modes: highly correlated branches, an unreliable source-trained ranking, or transformations that do not preserve the target label can remove the expected benefit.

S-II. ROBUSTNESS OF THE RELIABILITY-AWARE AGGREGATION

Two design choices determine whether BFT helps, namely the diversity of the transformation bank and the quality of the learned ranking. This section reports two analyses that probe each choice.

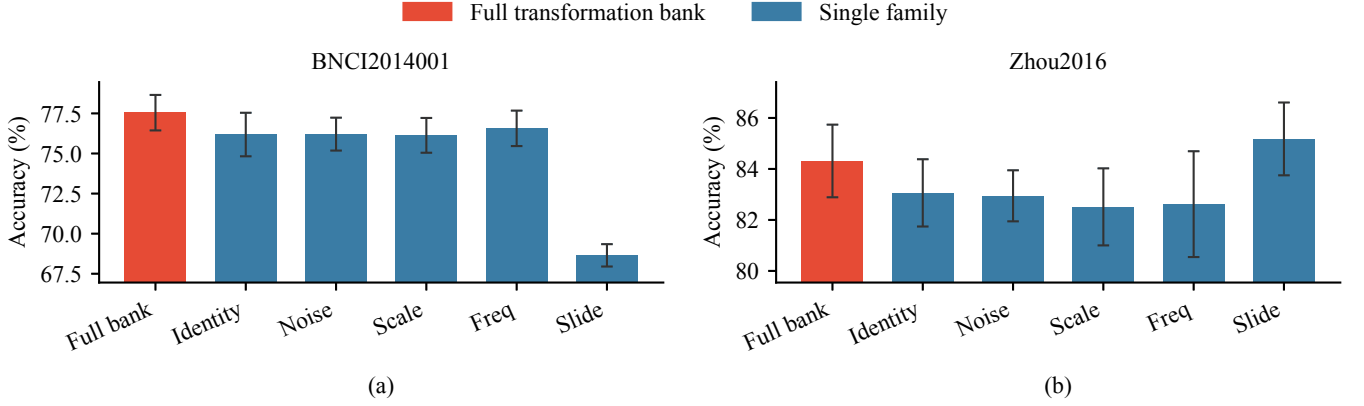


Fig. S2. Classification accuracy of the full BFT-A transformation bank compared with each single transformation family, on (a) BNCI2014001 and (b) Zhou2016. No single family is best on both datasets, so the full bank is the robust default.

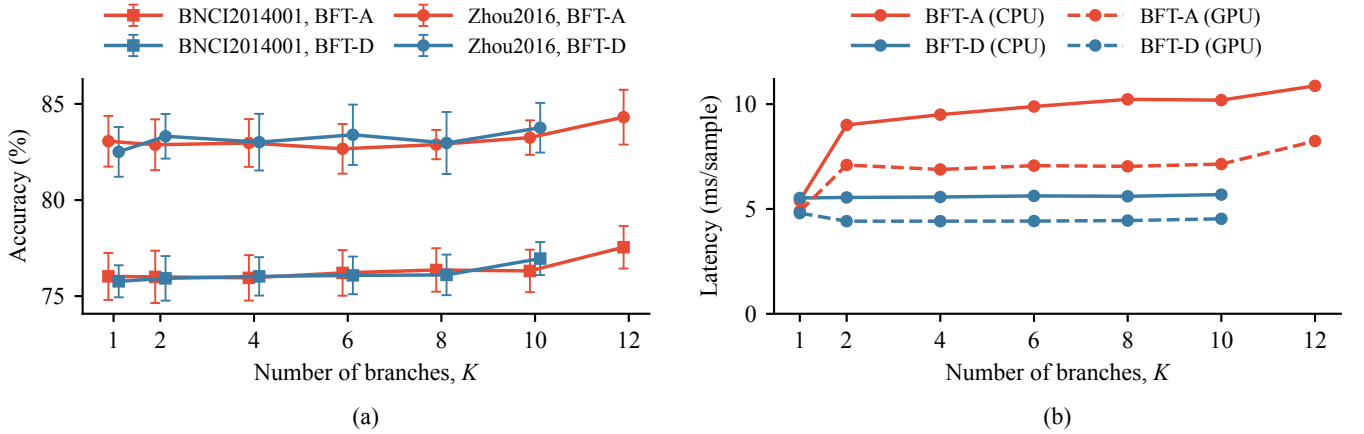


Fig. S3. Performance w.r.t. the number of branches K in BFT, with EEGNet. Error bars in (a) are standard deviations over the seed-subject units. BFT-D is swept up to its operating point $K = 10$, and BFT-A up to $K = 12$. (a) Accuracy; and (b) Latency.

A. Transformation Coverage

BFT aggregates branches produced by a bank of transformation families, so a natural concern is whether the method depends on any single family. Fig. S2 compares the full BFT-A bank against each single family on BNCI2014001 and Zhou2016. On BNCI2014001 the full bank reaches 77.55%, whereas the sliding-window family alone collapses to 68.65%. On Zhou2016 the sliding-window family alone is the strongest single choice at 85.18%, yet it is the weakest on BNCI2014001, and no single family is best on both datasets. The full bank is therefore the robust default, because it does not rely on any one family suiting the deployed data.

Varying the number of branches K showed that accuracy is not monotonic in K , as additional branches help only when they add useful views, as evident in Fig. S3. The operating points $K = 12$ for BFT-A and $K = 10$ for BFT-D gave the strongest measured accuracy for each variant under this protocol.

TABLE S1

EFFECT OF DEGRADING THE LEARNED RANKING AT TEST TIME ON CLASSIFICATION ACCURACY (%), REPLACING IT BY GAUSSIAN-PERTURBED, RANDOM, REVERSED, OR SHUFFLED RANKINGS. MEANS $\pm$ STANDARD DEVIATIONS OVER THE SEED-SUBJECT UNITS (27 FOR BNCI2014001, 12 FOR ZHOU2016). THE LEARNED RANKING IS BEST IN EVERY CASE.

Ranking	BNCI2014001		Zhou2016	
	BFT-A	BFT-D	BFT-A	BFT-D
Learned	77.52 $\pm$ 1.09	77.01 $\pm$ 0.83	84.31 $\pm$ 1.43	83.76 $\pm$ 1.29
Gaussian noise	75.75 $\pm$ 0.96	75.87 $\pm$ 1.23	83.41 $\pm$ 1.17	82.52 $\pm$ 1.30
Random	75.85 $\pm$ 1.36	76.16 $\pm$ 1.04	83.54 $\pm$ 1.41	82.73 $\pm$ 1.07
Reverse	75.46 $\pm$ 1.09	75.75 $\pm$ 0.96	83.01 $\pm$ 1.23	82.56 $\pm$ 1.18
Shuffle	76.24 $\pm$ 1.12	76.18 $\pm$ 1.18	83.73 $\pm$ 1.22	82.96 $\pm$ 1.44

B. Ranking Robustness

The reliability ranking is trained only on source subjects, so it is important to confirm that the ranking itself, and not the averaging alone, drives the improvement at test time. We replace the learned ranking with four degraded rankings, namely Gaussian-perturbed, random, reversed, and shuffled,

and measure the resulting accuracy. Table S1 shows that the learned ranking is best in every dataset and variant, and that every degraded ranking lowers accuracy, by up to 2.06% on BNCI2014001 for BFT-A when the ranking is reversed. The reliability structure learned on the source subjects therefore remains informative after a subject shift, and the ranking, not the uniform average, is responsible for the gain.

C. Regression Aggregation Rule

TABLE S2

REGRESSION AGGREGATION RULES ON THE DRIVING DATASET, AS THE CHANGE IN RMSE AND CC RELATIVE TO THE TOP-HALF MEAN USED IN THE MAIN TEXT. EVERY ALTERNATIVE STAYS WITHIN 0.005 RMSE AND 0.003 CC OF IT, FAR BELOW THE ACROSS-SEED STANDARD DEVIATION. AVERAGED OVER THREE SEEDS AND ALL SUBJECTS.

Aggregation rule	BFT-A		BFT-D	
	Δ RMSE	Δ CC	Δ RMSE	Δ CC
Top-half mean (used)	0.0000	0.0000	0.0000	0.0000
Reliability-weighted, all branches	-0.0002	+0.0030	+0.0019	+0.0013
Reliability-weighted, top half	0.0000	0.0000	0.0000	0.0000
Median	-0.0005	+0.0012	+0.0046	+0.0013
Trimmed mean	-0.0003	+0.0024	+0.0046	+0.0015

For regression, BFT averages the top-ranked half of the transformed predictions (Section III-D of the main text). We tested whether this discards useful information carried by the continuous reliability scores. On the Driving dataset we therefore compared the top-half mean against alternatives that do use the continuous scores or a different pooling rule, namely a reliability-weighted mean over all branches, a reliability-weighted mean over the top half, the median, and a trimmed mean. Table S2 reports each rule as the change in RMSE and CC relative to the top-half mean, so a value near zero means the rule matches it. Every alternative, including the continuous reliability weighting, stays within 0.005 in RMSE and 0.003 in CC of the top-half mean for both BFT-A and BFT-D. These differences are far smaller than the across-seed standard deviation, which is about 0.010 in RMSE and 0.017 in CC, so no rule is reliably better. This matches the design: the ranking objective supervises only the order of the branches, not calibrated inverse-error magnitudes, so the continuous scores carry no additional accuracy for continuous outputs, and the tuning-free top-half mean is preferred.